Benchmark Requirements

Domain: Overleaf / LaTeX (academic authoring)
Asset: Soil Health Proceedings — Owner: Rotimi Alade

Task

Prepare the *Proceedings of the International Symposium on Soil Health and Sustainable Crop Production, Volume 1* for publication by resolving the six editor's queries recorded in `notes/review_log.md`, working only in the Overleaf editor.

Required project content

- A proceedings volume with front matter, six contributed papers and back matter (List of Abbreviations and subject index).
- Each paper with title, abstract, keywords, introduction, materials and methods, results, discussion, conclusion and its own reference list.
- A synthetic, seeded dataset of 228 plots in five field trials at three sites, obeying 14 written data rules.
- Every table, figure and quoted number derived from the data through the results files; 25 qualitative statements consistent with the data.
- 18 references to published literature, each verified before use.

LaTeX requirements

- The project remains editable LaTeX source.
- Papers are connected to `main.tex` with `subfiles`; each also compiles on its own.
- The volume compiles with pdfLaTeX, biber and makeindex with 0 errors, 0 warnings and 0 overfull or underfull boxes.
- Existing references are used consistently; no references are added.

Acceptance thresholds

- Verifier score exactly 1.0000 (`verifier/verify.py`): all six defects fixed and all seven gate checks passed.
- Page count 40–42; 64 untouched files identical to the golden solution.
- Land equivalent ratio of S3 in the Paper 3 conclusion equal to the computed 1.37 within ± 0.005.
- All 9 columns of the sites table kept; no warning suppressed.
- Decoys `data/archive/t2_nitrogen_2025_export.tsv` and `bib/soilhealth_draft.bib` unchanged and unused.

Expected final state

The compiled volume of `golden_solution/`, with the milestone history kept in `milestone_1_foundation/` to `milestone_6_verification/` and `CHECKPOINTS.md`.

Important: the symposium is fictional and the dataset and results are synthetic and demonstrative. They must not be presented as evidence about real soils, crops or farming systems.